

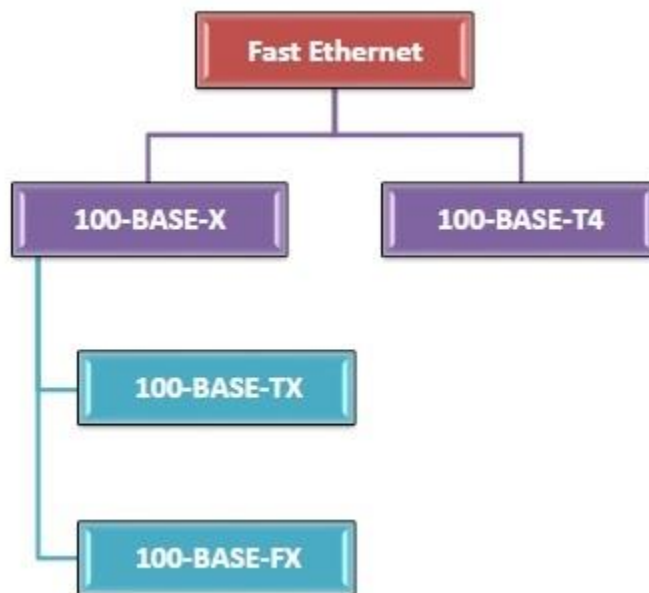
Fast Ethernet (802.3u)

In computer networks, Fast Ethernet is a variation of Ethernet standards that carry data traffic at 100 Mbps (Mega bits per second) in local area networks (LAN). It was launched as the IEEE 802.3u standard in 1995, and stayed the fastest network till the introduction of Gigabit Ethernet.

Fast Ethernet is popularly named as 100-BASE-X. Here, 100 is the maximum throughput, i.e. 100 Mbps, BASE denoted use of baseband transmission, and X is the type of medium used, which is TX or FX.

Varieties of Fast Ethernet

The common varieties of fast Ethernet are 100-Base-TX, 100-BASE-FX and 100-Base-T4.



- **100-Base-T4**
 - This has four pairs of UTP of Category 3, two of which are bi-directional and the other two are unidirectional.
 - In each direction, three pairs can be used simultaneously for data transmission.
 - Each twisted pair is capable of transmitting a maximum of 25Mbaud data. Thus the three pairs can handle a maximum of 75Mbaud data.
 - It uses the encoding scheme 8B/6T (eight binary/six ternary).
- **100-Base-TX**

- This has either two pairs of unshielded twisted pairs (UTP) category 5 wires or two shielded twisted pairs (STP) type 1 wires. One pair transmits frames from hub to the device and the other from device to hub.
- Maximum distance between hub and station is 100m.
- It has a data rate of 125 Mbps.
- It uses MLT-3 encoding scheme along with 4B/5B block coding.
- **100-BASE-FX**
 - This has two pairs of optical fibers. One pair transmits frames from hub to the device and the other from device to hub.
 - Maximum distance between hub and station is 2000m.
 - It has a data rate of 125 Mbps.
 - It uses NRZ-I encoding scheme along with 4B/5B block coding.

Frame Format of IEEE 802.3

The frame format of IEEE 802.3u is same as IEEE 802.3. The fields in the frame are:

- **Preamble** – It is a 7 bytes starting field that provides alert and timing pulse for transmission.
- **Start of Frame Delimiter (SOF)** – It is a 1 byte field that contains an alternating pattern of ones and zeros ending with two ones.
- **Destination Address** – It is a 6 byte field containing physical address of destination stations.
- **Source Address** – It is a 6 byte field containing the physical address of the sending station.
- **Length** – It is a 2 bytes field that stores the number of bytes in the data field.
- **Data** – This is a variable sized field carries the data from the upper layers. The maximum size of data field is 1500 bytes.
- **Padding** – This is added to the data to bring its length to the minimum requirement of 46 bytes.
- **CRC** – CRC stands for cyclic redundancy check. It contains the error detection information.

Name	Cable	Max. segment	Advantages
100Base-T4	Twisted pair	100 m	Uses category 3 UTP
100Base-TX	Twisted pair	100 m	Full duplex at 100 Mbps (Cat 5 UTP)
100Base-FX	Fiber optics	2000 m	Full duplex at 100 Mbps; long runs

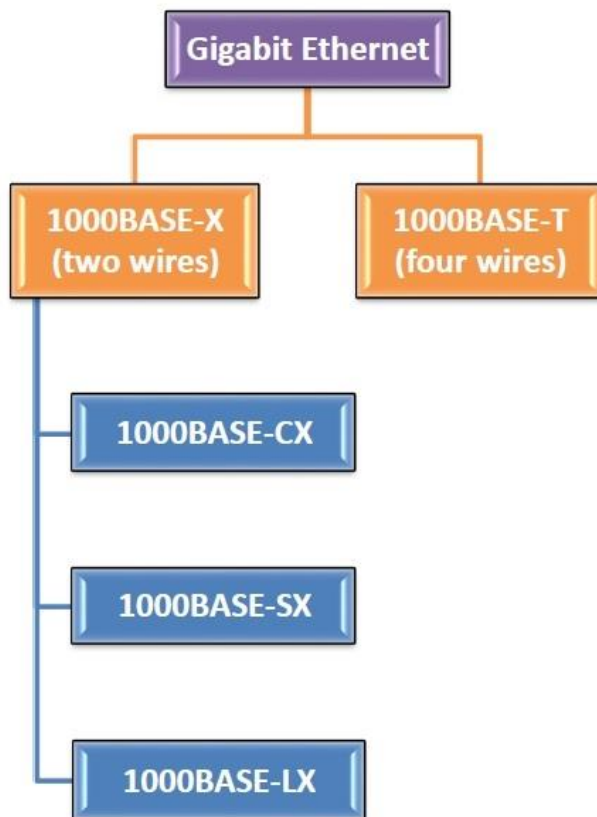
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Gigabit Ethernet

In computer networks, Gigabit Ethernet (GbE) is the family of Ethernet technologies that achieve theoretical data rates of 1 gigabit per second (1 Gbps). It was introduced in 1999 and was defined by the IEEE 802.3ab standard.

Varieties of Gigabit Ethernet

The popular varieties of fast Ethernet are 1000Base-SX, 1000Base-LX, 1000BASE-T and 1000Base-CX.



1000BASE-CX

- Defined by IEEE 802.3z standard
- The initial standard for Gigabit Ethernet
- Uses shielded twisted pair cables with DE-9 or 8P8C connector
- Maximum segment length is 25 metres
- Uses NRZ line encoding and 8B/6B block encoding

1000BASE-SX

- Defined by IEEE 802.3z standard
- Uses a pair of fibre optic cables of a shorter wavelength having 770 – 860 nm diameter
- The maximum segment length varies from 220 – 550 metres depending upon the fiber properties.
- Uses NRZ line encoding and 8B/10B block encoding

1000BASE-LX

- Defined by IEEE 802.3z standard
- Uses a pair of fibre optic cables of a longer wavelength having 1270 – 1355 nm diameter
- Maximum segment length is 500 metres
- Can cover distances up to 5 km
- Uses NRZ line encoding and 8B/10B block encoding

1000BASE-T

- Defined by IEEE 802.3ab standard
- Uses a pair four lanes of twisted-pair cables (Cat-5, Cat-5e, Cat-6, Cat-7)
- Maximum segment length is 100 metres
- Uses trellis code modulation technique

Name	Cable	Max. segment	Advantages
1000Base-SX	Fiber optics	550 m	Multimode fiber (50, 62.5 microns)
1000Base-LX	Fiber optics	5000 m	Single (10 μ) or multimode (50, 62.5 μ)
1000Base-CX	2 Pairs of STP	25 m	Shielded twisted pair
1000Base-T	4 Pairs of UTP	100 m	Standard category 5 UTP

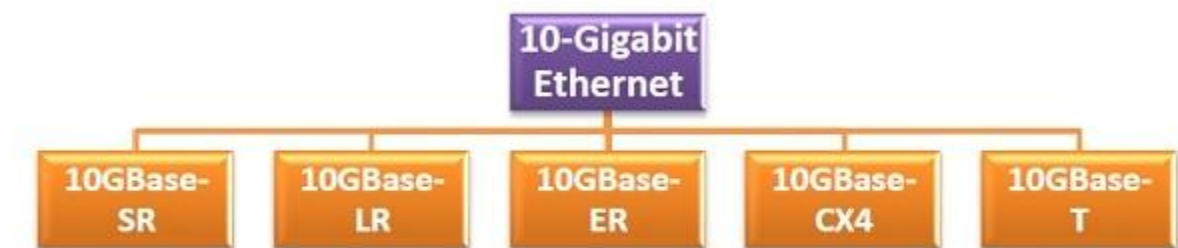
10-Gigabit Ethernet

In computer networks, 10-Gigabit Ethernet is the family of Ethernet technologies that achieve maximum rates up to 10 gigabits per second (10 Gbps). It is also known as 10GE, 10GbE or 10 GigE. It is defined by the IEEE 802.3ae-2002 standard.

10GE is a thousand times faster than standard Ethernet and supports only full-duplex communication. Multimode fiber having 0.85 μ frequency is used for medium distances and single-mode fiber having 1.5 μ frequency is used for long distances.

Varieties of Gigabit Ethernet

The popular varieties of fast Ethernet are 1000Base-SX, 1000Base-LX, 1000BASE-T and 1000Base-CX.



10GBase-SR

- Defined by IEEE 802.3ae standard
- Uses fiber optic cables
- Maximum segment length is 300 m
- Deployed using multimode fibers having 0.85 μ frequency

10GBase-LR

- Defined by IEEE 802.3ae standard
- Uses fiber optic cables
- Maximum segment length is 10 km
- Deployed using single-mode fibers having 1.3 μ frequency

10GBase-ER

- Defined by IEEE 802.3ae standard
- Uses fiber optic cables
- Maximum segment length is 40 km
- Deployed using single-mode fibers having 1.5 μ frequency

10GBase-CX4

- Defined by IEEE 802.3ak standard
- Uses 4 pairs of twin-axial cables
- Maximum segment length is 15 m

- Uses 8B/10B coding

10GBase-T

- Defined by IEEE 802.3an standard
- Uses 4 pairs of unshielded twisted pair cables
- Maximum segment length is 100 m
- Uses low-density parity-check code (LDPC code)

Name	Cable	Max. segment	Advantages
10GBase-SR	Fiber optics	Up to 300 m	Multimode fiber (0.85μ)
10GBase-LR	Fiber optics	10 km	Single-mode fiber (1.3μ)
10GBase-ER	Fiber optics	40 km	Single-mode fiber (1.5μ)
10GBase-CX4	4 Pairs of twinax	15 m	Twinaxial copper
10GBase-T	4 Pairs of UTP	100 m	Category 6a UTP